



Deepsikha Pramanick

Roll No.: 24035010521

Online BSc (Hons.) - Data Science and AI

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EDUCATION

Degree Discipline	Institute/Board	CGPA/Percentage	Year
BSc (Hons.) Data Science and AI	Indian Institute of Technology, Guwahati	7.53	2024-Present
Senior Secondary	Guptipara High School(H.S.)/ WBCHSE	89.4 %	2019
Secondary	Guptipara Girls' High School(H.S.)/ WBBSE	94.86 %	2017

EXPERIENCE

- Research Project — BSA-Integrated Chitosan Hydrogels** 2026 – Present
Student Researcher IIT Guwahati
 - Developing an computational workflow integrating molecular docking, molecular dynamics, machine learning and literature-derived experimental data for controlled drug release and wound-healing biomaterials.
 - Built and analyzed a structured literature dataset on chitosan-based hydrogels and investigated formulation, processing and release-related factors for data-driven modelling.
- Research Project — M. tuberculosis Hsp16.3** 2026
Student Researcher IIT Guwahati / IIT Bhubaneswar
 - Studied site-specific acetylation of Mycobacterium tuberculosis Hsp16.3 using molecular simulation and machine-learning-based analysis.
 - Contributed as second author to a research paper examining the relationship between acetylation and Hsp16.3 chaperone efficiency.

PROJECTS

- RNA Structure Prediction & Rapid Antiviral Drug Development** 2025 – Present
MFCC CARE Conference, IIT Guwahati GitHub
 - Developed an AI-driven computational workflow for RNA 3D structure prediction and rapid antiviral drug discovery.
 - Explored computational structure prediction, molecular modelling and drug-design workflows for biomolecular systems.
- AI-Guided Design of BSA-Integrated Chitosan Hydrogels** 2026 – Present
IIT Guwahati
 - Integrating molecular simulation, machine learning and experimental literature analysis to optimize controlled drug release and wound-healing biomaterials.
 - Developing and analyzing structured datasets and predictive models for formulation and controlled drug-release.
- Site-Specific Acetylation of Mycobacterium tuberculosis Hsp16.3** 2026
Computational Research Project
 - Investigated the effect of site-specific lysine acetylation on the structural and dynamic behaviour of Mycobacterium tuberculosis Hsp16.3.
 - Performed molecular dynamics analysis using GROMACS on HPC PARAM KAMRUPA, including trajectory-based RMSD, RMSF, PCA and free-energy landscape and ML analysis.

TECHNICAL SKILLS

- Programming:** Python, C, C#, R, Java, SQL, Bash, HTML, CSS, JavaScript
- Data Science & AI:** NumPy, Pandas, SciPy, scikit-learn, PyTorch, Jupyter, Git
- Molecular Simulation:** GROMACS, CHARMM-GUI, AMBER, Molecular Dynamics, Molecular Docking
- Scientific Computing & HPC:** PARAM Kamrupa, HPC, Linux, Bash Scripting
- Computational Chemistry:** Avogadro, BURAI, VESTA, Quantum ESPRESSO, JDK
- Engineering Design:** SolidWorks, Fusion 360, AutoCAD
- Data Visualization:** Matplotlib, Power BI, Tableau

KEY COURSES TAKEN

- Data Science & AI:** Machine Learning Fundamentals, Recommender Systems, Time Series Analysis & Forecasting, Data Structure, Algorithm, Data Modeling & Visualization
- Programming:** Python Programming, R, C, Java
- Signals & Systems:** Signals and Systems

POSITIONS OF RESPONSIBILITY

ACHIEVEMENTS

- **Presented research poster at MFF CARE Conference**, Mehta Family School of Data Science & AI, IIT Guwahati *2025*
- **Presented LOCUS project at One Health Ideathon**, JBMSHST, IIT Guwahati *2026*
- **Co-authored first research paper**, IIT Guwahati / IIT Bhubaneswar *2026*